

[Home](#) / [Class - 8 Concise Mathematics Selina](#) / [Identities](#)

Chapter 12

Identities

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STUDY MATERIAL ▾

LOGIN

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Search by lesson title

Chapter 11
Algebraic Expressions

Chapter 12
Identities

Chapter 13
Factorisation

Chapter 14
Linear Equations in One Variable

Chapter 15
Linear Inequalities

Chapter 16
Understanding Shapes

Chapter 17
Special Types of Quadrilaterals

Chapter 18
Constructions

Chapter 19
Representing 3-D in 2-D

Biological Sciences

Government

Vocational & Continuing Education



CONTENTS

[Exercise 12\(A\)](#)[Exercise 12\(B\)](#)[Test Yourself](#)

Exercise 12(A)

Question 1(i)

 $(x + 2y)^2 + (x - 2y)^2$ is equal to:

1. $2x^2 + 8y^2 + 8xy$

2. $x^2 + 4y^2$

3. $2x^2 + 8y^2$



$$4. 2x^2 - 8y^2$$

Answer

$$(x + 2y)^2 + (x - 2y)^2$$

And using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

and,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$= [x^2 + 2 \times x \times 2y + (2y)^2] + [x^2 - 2 \times x \times 2y + (2y)^2]$$

$$= [x^2 + 4xy + 4y^2] + [x^2 - 4xy + 4y^2]$$

$$= x^2 + 4xy + 4y^2 + x^2 - 4xy + 4y^2$$

$$= (x^2 + x^2) + (4xy - 4xy) + (4y^2 + 4y^2)$$

$$= 2x^2 + 8y^2$$

Hence, option 3 is the correct option.

Question 1(ii)

$(a + b)(a - b) + (b - c)(b + c) + (c + a)(c - a)$ is equal to:

$$1. 2a^2 + 2b^2 + 2c^2$$

$$2. a^2 + b^2 + c^2 - 2ab - 2bc - 2ca$$

3. 0

4. none of these

Answer

$$[(a + b)(a - b)] + [(b - c)(b + c)] + [(c + a)(c - a)]$$

Using the formula,

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= [a^2 - b^2] + [b^2 - c^2] + [c^2 - a^2]$$

$$= a^2 - b^2 + b^2 - c^2 + c^2 - a^2$$

$$= (a^2 - a^2) + (-b^2 + b^2) + (-c^2 + c^2)$$

$$= 0$$

Hence, option 3 is the correct option.

Question 1(iii)

$(3x - 4y)^2 - (3x + 4y)^2$ is equal to:

1. $18x^2 + 32y^2$

2. $18x^2 - 32y^2$

3. $-48xy$

4. $48xy$

Answer

$$(3x - 4y)^2 - (3x + 4y)^2$$

Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

and,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$= [(3x)^2 - 2 \times (3x) \times (4y) + (4y)^2] - [(3x)^2 + 2 \times (3x) \times (4y) + (4y)^2]$$

$$= [9x^2 - 24xy + 16y^2] - [9x^2 + 24xy + 16y^2]$$

$$= 9x^2 - 24xy + 16y^2 - 9x^2 - 24xy - 16y^2$$

$$= (9x^2 - 9x^2) + (-24xy - 24xy) + (16y^2 - 16y^2)$$

$$= -48xy$$

Hence, option 3 is the correct option.

Question 1(iv)

The value of $(0.8)^2 - 0.32 + (0.2)^2$ is equal to:

1. 1

2. 3.6

3. 0.36

4. 0.036

Answer

$$(0.8)^2 - 0.32 + (0.2)^2$$

$$= (0.8)^2 - 2 \times (0.8) \times (0.2) + (0.2)^2$$

Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$= [0.8 - 0.2]^2$$

$$= 0.6^2$$

$$= 0.36$$

Hence, option 3 is the correct option.**Question 1(v)**The value of $(a - b - c)(a - b + c)$ is:

1. $a^2 + b^2 + c^2 + 2ab$

2. $a^2 + b^2 - c^2 - 2ab$

3. $a^2 + b^2 + c^2 - 2ab$

4. $a^2 + b^2 - c^2 + 2ab$

Answer

$$\begin{aligned}& (a - b - c)(a - b + c) \\&= a(a - b + c) - b(a - b + c) - c(a - b + c) \\&= a^{(1+1)} - ab + ac - ab + b^{(1+1)} - bc - ac + bc - c^{(1+1)} \\&= a^2 + (-ab - ab) + (ac - ac) + b^2 + (-bc + bc) - c^2 \\&= a^2 - 2ab + b^2 - c^2\end{aligned}$$

Hence, option 2 is the correct option.

Question 2(i)

Use direct method to evaluate the following product:

$$(a - 8)(a + 2)$$

Answer

$$(a - 8)(a + 2)$$

Using the formula,

$$(x - a)(x + b) = x^2 - (a - b)x - ab$$

$$= a^2 - (8 - 2)a - 8 \times 2$$

$$= a^2 - 6a - 16$$

$$\text{Hence, } (a - 8)(a + 2) = a^2 - 6a - 16$$

Question 2(ii)

Use direct method to evaluate the following product:

$$(b - 3)(b - 5)$$

Answer

$$(b - 3)(b - 5)$$

Using the formula,

$$(x + a)(x + b) = x^2 - (a + b)x + ab$$

$$= b^2 - (3 + 5)b + 3 \times 5$$

$$= b^2 - 8b + 15$$

$$\text{Hence, } (b - 3)(b - 5) = b^2 - 8b + 15$$

Question 2(iii)

Use direct method to evaluate the following product:

$$(3x - 2y)(2x + y)$$

Answer

$$(3x - 2y)(2x + y)$$

$$= (3x \times 2x) + (3x \times y + (-2y) \times 2x) + ((-2y) \times y)$$

$$= 6x^2 + (3xy - 4xy) - 2y^2$$

$$= 6x^2 - 1xy - 2y^2$$

Hence, $(3x - 2y)(2x + y) = 6x^2 - xy - 2y^2$

Question 2(iv)

Use direct method to evaluate the following product:

$$(5a + 16)(3a - 7)$$

Answer

$$(5a + 16)(3a - 7)$$

$$= (5a \times 3a) + (5a \times (-7)) + (16 \times 3a) + (16 \times (-7))$$

$$= 15a^2 + (-35a + 48a) - 112$$

$$= 15a^2 + 13a - 112$$

Hence, $(5a + 16)(3a - 7) = 15a^2 + 13a - 112$

Question 2(v)

Use direct method to evaluate the following product:

$$(8 - b)(3 + b)$$

Answer

$$(8 - b)(3 + b)$$

$$= (8 \times 3) + (8 \times b) + (-b \times 3) + ((-b) \times b)$$

$$= 24 + (8b - 3b) - b^2$$

$$= 24 + 5b - b^2$$

$$\text{Hence, } (8 - b)(3 + b) = 24 + 5b - b^2$$

Question 3(i)

Evaluate:

$$(2a + 3)(2a - 3)$$

Answer

$$(2a + 3)(2a - 3)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= (2a)^2 - 3^2$$

$$= 4a^2 - 9$$

$$\text{Hence, } (2a + 3)(2a - 3) = 4a^2 - 9$$

Question 3(ii)

Evaluate:

$$(xy + 4)(xy - 4)$$

Answer

$$(xy + 4)(xy - 4)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= (xy)^2 - 4^2$$

$$= x^2y^2 - 16$$

$$\text{Hence, } (xy + 4)(xy - 4) = x^2y^2 - 16$$

Question 3(iii)

Evaluate:

$$(ab + x^2)(ab - x^2)$$

Answer

$$(ab + x^2)(ab - x^2)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= (ab)^2 - (x^2)^2$$

$$= a^2b^2 - x^4$$

$$\text{Hence, } (ab + x^2)(ab - x^2) = a^2b^2 - x^4$$

Question 3(iv)

Evaluate:

$$(3x^2 + 5y^2)(3x^2 - 5y^2)$$

Answer

$$(3x^2 + 5y^2)(3x^2 - 5y^2)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= (3x^2)^2 - (5y^2)^2$$

$$= 9x^4 - 25y^4$$

$$\text{Hence, } (3x^2 + 5y^2)(3x^2 - 5y^2) = 9x^4 - 25y^4$$

Question 3(v)

Evaluate:

$$\left(z - \frac{2}{3}\right) \left(z + \frac{2}{3}\right)$$

Answer

$$\left(z - \frac{2}{3}\right) \left(z + \frac{2}{3}\right)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= z^2 - \left(\frac{2}{3}\right)^2$$

$$= z^2 - \frac{4}{9}$$

$$\text{Hence, } \left(z - \frac{2}{3}\right) \left(z + \frac{2}{3}\right) = z^2 - \frac{4}{9}$$

Question 3(vi)

Evaluate:

$$\left(\frac{3}{5}a + \frac{1}{2}\right) \left(\frac{3}{5}a - \frac{1}{2}\right)$$

Answer

$$\left(\frac{3}{5}a + \frac{1}{2}\right) \left(\frac{3}{5}a - \frac{1}{2}\right)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= \left(\frac{3}{5}a\right)^2 - \left(\frac{1}{2}\right)^2$$

$$= \left(\frac{9}{25}a^2\right) - \frac{1}{4}$$

$$\text{Hence, } \left(\frac{3}{5}a + \frac{1}{2}\right) \left(\frac{3}{5}a - \frac{1}{2}\right) = \left(\frac{9}{25}a^2\right) - \frac{1}{4}$$

Question 3(vii)

Evaluate:

$$(0.5 - 2a)(0.5 + 2a)$$

Answer

$$(0.5 - 2a)(0.5 + 2a)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= (0.5)^2 - (2a)^2$$

$$= 0.25 - 4a^2$$

$$\text{Hence, } (0.5 - 2a)(0.5 + 2a) = 0.25 - 4a^2$$

Question 3(viii)

Evaluate:

$$\left(\frac{a}{2} - \frac{b}{3}\right) \left(\frac{a}{2} + \frac{b}{3}\right)$$

Answer

$$\left(\frac{a}{2} - \frac{b}{3}\right) \left(\frac{a}{2} + \frac{b}{3}\right)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= \left(\frac{a}{2}\right)^2 - \left(\frac{b}{3}\right)^2$$

$$= \left(\frac{a^2}{4}\right) - \left(\frac{b^2}{9}\right)$$

$$\text{Hence, } \left(\frac{a}{2} - \frac{b}{3}\right) \left(\frac{a}{2} + \frac{b}{3}\right) = \left(\frac{a^2}{4}\right) - \left(\frac{b^2}{9}\right)$$

Question 4(i)

Evaluate:

$$(a + b) (a - b) (a^2 + b^2)$$

Answer

$$(a + b) (a - b) (a^2 + b^2)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= (a^2 - b^2) (a^2 + b^2)$$

$$= a^4 - b^4$$

$$\text{Hence, } (a + b) (a - b) (a^2 + b^2) = a^4 - b^4$$

Question 4(ii)

Evaluate:

$$(3 - 2x) (3 + 2x) (9 + 4x^2)$$

Answer

$$(3 - 2x) (3 + 2x) (9 + 4x^2)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= (3^2 - (2x)^2) (9 + 4x^2)$$

$$= (9 - 4x^2) (9 + 4x^2)$$

$$= 9^2 - (4x^2)^2$$

$$= 81 - 16x^4$$

$$\text{Hence, } (3 - 2x) (3 + 2x) (9 + 4x^2) = 81 - 16x^4$$

Question 4(iii)

Evaluate:

$$(3x - 4y) (3x + 4y) (9x^2 + 16y^2)$$

Answer

$$(3x - 4y) (3x + 4y) (9x^2 + 16y^2)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= [(3x)^2 - (4y)^2] (9x^2 + 16y^2)$$

$$= (9x^2 - 16y^2) (9x^2 + 16y^2)$$

$$= (9x^2)^2 - (16y^2)^2$$

$$= 81x^4 - 256y^4$$

$$\text{Hence, } (3x - 4y) (3x + 4y) (9x^2 + 16y^2) = 81x^4 - 256y^4$$

Question 5

Use the formula: $(a + b)(a - b) = a^2 - b^2$ to evaluate:

(i) 21×19

(ii) 33×27

(iii) 103×97

(iv) 9.8×10.2

(v) 7.7×8.3

Answer

(i) 21×19

$(20 + 1)(20 - 1)$

Using the formula: $(a + b)(a - b) = a^2 - b^2$

$$\Rightarrow (20 + 1)(20 - 1) = 20^2 - 1^2$$

$$= 400 - 1$$

$$= 399$$

Hence, $21 \times 19 = 399$.

(ii) 33×27

$(30 + 3)(30 - 3)$

Using the formula: $(a + b)(a - b) = a^2 - b^2$

$$\Rightarrow (30 + 3)(30 - 3) = 30^2 - 3^2$$

$$= 900 - 9$$

$$= 891$$

Hence, $33 \times 27 = 891$.

(iii) 103×97

$$(100 + 3)(100 - 3)$$

Using the formula: $(a + b)(a - b) = a^2 - b^2$

$$\Rightarrow (100 + 3)(100 - 3) = 100^2 - 3^2$$

$$= 10000 - 9$$

$$= 9991$$

Hence, $103 \times 97 = 9991$.

(iv) 9.8×10.2

$$(10 - 0.2)(10 + 0.2)$$

Using the formula: $(a + b)(a - b) = a^2 - b^2$

$$\Rightarrow (10 - 0.2)(10 + 0.2) = 10^2 - (0.2)^2$$

$$= 100 - 0.04$$

$$= 99.96$$

Hence, $9.8 \times 10.2 = 99.96$.

$$(v) 7.7 \times 8.3$$

$$(8 - 0.3)(8 + 0.3)$$

Using the formula: $(a + b)(a - b) = a^2 - b^2$

$$\Rightarrow (8 - 0.3)(8 + 0.3) = 8^2 - (0.3)^2$$

$$= 64 - 0.09$$

$$= 63.91$$

Hence, $7.7 \times 8.3 = 63.91$.

Question 6(i)

Evaluate:

$$(6 - xy)(6 + xy)$$

Answer

$$(6 - xy)(6 + xy)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= 6^2 - (xy)^2$$

$$= 36 - x^2y^2$$

Hence, $(6 - xy)(6 + xy) = 36 - x^2y^2$

Question 6(ii)

Evaluate:

$$\left(7x + \frac{2}{3}y\right) \left(7x - \frac{2}{3}y\right)$$

Answer

$$\left(7x + \frac{2}{3}y\right) \left(7x - \frac{2}{3}y\right)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= \left(7x\right)^2 - \left(\frac{2}{3}y\right)^2$$

$$= 49x^2 - \frac{4}{9}y^2$$

$$\text{Hence, } \left(7x + \frac{2}{3}y\right) \left(7x - \frac{2}{3}y\right) = 49x^2 - \left(\frac{4}{9}y^2\right)$$

Question 6(iii)

Evaluate:

$$\left(\frac{a}{2b} + \frac{2b}{a}\right) \left(\frac{a}{2b} - \frac{2b}{a}\right)$$

Answer

$$\left(\frac{a}{2b} + \frac{2b}{a}\right) \left(\frac{a}{2b} - \frac{2b}{a}\right)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= \left(\frac{a}{2b}\right)^2 - \left(\frac{2b}{a}\right)^2$$

$$= \left(\frac{a^2}{4b^2}\right) - \left(\frac{4b^2}{a^2}\right)$$

$$\text{Hence, } \left(\frac{a}{2b} + \frac{2b}{a}\right) \left(\frac{a}{2b} - \frac{2b}{a}\right) = \left(\frac{a^2}{4b^2}\right) - \left(\frac{4b^2}{a^2}\right)$$

Question 6(iv)

Evaluate:

$$\left(3x - \frac{1}{2y}\right) \left(3x + \frac{1}{2y}\right)$$

Answer

$$\left(3x - \frac{1}{2y}\right) \left(3x + \frac{1}{2y}\right)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= (3x)^2 - \left(\frac{1}{2y}\right)^2$$

$$= 9x^2 - \left(\frac{1}{4y^2}\right)$$

$$\text{Hence, } \left(3x - \frac{1}{2y}\right) \left(3x + \frac{1}{2y}\right) = 9x^2 - \left(\frac{1}{4y^2}\right)$$

Question 6(v)

Evaluate:

$$(2a + 3) (2a - 3) (4a^2 + 9)$$

Answer

$$(2a + 3) (2a - 3) (4a^2 + 9)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= [(2a)^2 - 3^2] (4a^2 + 9)$$

$$= (4a^2 - 9^2) (4a^2 + 9)$$

$$= (4a^2)^2 - 9^2$$

$$= 16a^4 - 81$$

$$\text{Hence, } (2a + 3) (2a - 3) (4a^2 + 9) = 16a^4 - 81$$

Question 6(vi)

Evaluate:

$$(a + bc) (a - bc) (a^2 + b^2c^2)$$

Answer

$$(a + bc) (a - bc) (a^2 + b^2c^2)$$

Using the formula

$$[\because (x + y)(x - y) = x^2 - y^2]$$

$$= [a^2 - (bc)^2] (a^2 + b^2c^2)$$

$$= (a^2 - b^2c^2) (a^2 + b^2c^2)$$

$$= (a^2)^2 - (b^2c^2)^2$$

$$= a^4 - b^4c^4$$

$$\text{Hence, } (a + bc) (a - bc) (a^2 + b^2c^2) = a^4 - b^4c^4$$

Question 7(i)

Expand:

$$\left(a + \frac{1}{2a}\right)^2$$

Answer

$$\left(a + \frac{1}{2a}\right)^2$$

Using the formula:

$$(\because (x + y)^2 = x^2 + 2xy + y^2)$$

$$= (a)^2 + 2 \times a \times \frac{1}{2a} + \left(\frac{1}{2a}\right)^2$$

$$= a^2 + \frac{2a}{2a} + \frac{1}{4a^2}$$

$$= a^2 + 1 + \frac{1}{4a^2}$$

Hence, $\left(a + \frac{1}{2a}\right)^2 = a^2 + 1 + \left(\frac{1}{4a^2}\right)$

Question 7(ii)

Expand:

$$\left(2a - \frac{1}{a}\right)^2$$

Answer

$$\left(2a - \frac{1}{a}\right)^2$$

Using the formula:

$$(\because (x + y)^2 = x^2 + 2xy + y^2)$$

$$= (2a)^2 - 2 \times 2a \times \frac{1}{a} + \left(\frac{1}{a}\right)^2$$

$$= 4a^2 - \frac{4a}{a} + \frac{1}{a^2}$$

$$= 4a^2 - 4 + \frac{1}{a^2}$$

Hence, $\left(2a - \frac{1}{a}\right)^2 = 4a^2 - 4 + \left(\frac{1}{a^2}\right)$

Question 7(iii)

Expand:

$$(a + b - c)^2$$

Answer

$$(a + b - c)^2$$

Using the formula,

$$[\because (x + y - z)^2 = x^2 + y^2 + z^2 + 2xy - 2yz - 2xz]$$

$$= a^2 + b^2 + c^2 + 2ab - 2bc - 2ca$$

$$\text{Hence, } (a + b - c)^2 = a^2 + b^2 + c^2 + 2ab - 2bc - 2ca$$

Question 7(iv)

Expand:

$$(a - b + c)^2$$

Answer

$$(a - b + c)^2$$

Using the formula,

$$[\because (x - y + z)^2 = x^2 + y^2 + z^2 - 2xy - 2yz + 2xz]$$

$$= a^2 + b^2 + c^2 - 2ab - 2bc + 2ca$$

$$\text{Hence, } (a + b - c)^2 = a^2 + b^2 + c^2 - 2ab - 2bc + 2ca$$

Question 7(v)

Expand:

$$\left(3x + \frac{1}{3x}\right)^2$$

Answer

$$\left(3x + \frac{1}{3x}\right)^2$$

Using the formula

$$(\because (x + y)^2 = x^2 + 2xy + y^2)$$

$$= (3x)^2 + 2 \times 3x \times \frac{1}{3x} + \frac{1}{3x}^2$$

$$= 9x^2 + \frac{6x}{3x} + \frac{1}{9x^2}$$

$$= 9x^2 + 2 + \frac{1}{9x^2} \text{ Hence, } \left(3x + \frac{1}{3x}\right)^2 = 9x^2 + 2 + \frac{1}{9x^2}$$

Question 8(i)

Find the square of:

$$a + \frac{1}{5a}$$

Answer

$$\left(a + \frac{1}{5a}\right)^2$$

Using the formula

$$(\because (x + y)^2 = x^2 + 2xy + y^2)$$

$$= a^2 + 2 \times a \times \frac{1}{5a} + \frac{1}{5a}^2$$

$$= a^2 + \frac{2a}{5a} + \frac{1}{25a^2}$$

$$= a^2 + \frac{2}{5} + \frac{1}{25a^2}$$

$$\text{Hence, } \left(a + \frac{1}{5a}\right)^2 = a^2 + \frac{2}{5} + \frac{1}{25a^2}$$

Question 8(ii)

Find the square of:

$$2a - \frac{1}{a}$$

Answer

$$\left(2a - \frac{1}{a}\right)^2$$

Using the formula

$$(\because (x - y)^2 = x^2 - 2xy + y^2)$$

$$= (2a)^2 - 2 \times 2a \times \frac{1}{a} + \frac{1}{a}^2$$

$$= 4a^2 - \frac{4a}{a} + \frac{1}{a^2}$$

$$= 4a^2 - 4 + \frac{1}{a^2}$$

Hence, $\left(2a - \frac{1}{a}\right)^2 = 4a^2 - 4 + \frac{1}{a^2}$

Question 8(iii)

Find the square of:

$$x - 2y + 1$$

Answer

$$(x - 2y + 1)^2$$

Using the formula

$$(\because (x - y + z)^2 = x^2 + y^2 + z^2 - 2xy - 2yz + 2xz)$$

$$= x^2 + (2y)^2 + 1^2 - 2 \times (x) \times (2y) - 2 \times (2y) \times 1 + 2 \times 1 \times (x)$$

$$= x^2 + 4y^2 + 1 - 4xy - 4y + 2x$$

Hence, $(x - 2y + 1)^2 = x^2 + 4y^2 + 1 - 4xy - 4y + 2x$

Question 8(iv)

Find the square of:

$$3a - 2b - 5c$$

Answer

$$(3a - 2b - 5c)^2$$

Using the formula

$$\begin{aligned}
 (\because (x - y - z)^2 &= x^2 + y^2 + z^2 - 2xy + 2yz - 2xz) \\
 &= (3a)^2 + (2b)^2 + (5c)^2 - 2 \times (3a) \times (2b) - 2 \times (2b) \times (5c) + 2 \times (5c) \times (3a) \\
 &= 9a^2 + 4b^2 + 25c^2 - 12ab + 20bc - 30ca
 \end{aligned}$$

$$\text{Hence, } (3a - 2b - 5c)^2 = 9a^2 + 4b^2 + 25c^2 - 12ab + 20bc - 30ca$$

Question 8(v)

Find the square of:

$$2x + \frac{1}{x} + 1$$

Answer

$$\left(2x + \frac{1}{x} + 1\right)^2$$

Using the formula

$$\begin{aligned}
 (\because (x + y + z)^2 &= x^2 + y^2 + z^2 + 2xy + 2yz + 2xz) \\
 &= (2x)^2 + \left(\frac{1}{x}\right)^2 + 1^2 + 2 \times 2x \times \left(\frac{1}{x}\right) + 2 \times \left(\frac{1}{x}\right) \times 1 + 2 \times 1 \times 2x \\
 &= 4x^2 + \left(\frac{1}{x^2}\right) + 1 + \left(\frac{4x}{x}\right) + \left(\frac{2}{x}\right) + 4x \\
 &= 4x^2 + \left(\frac{1}{x^2}\right) + 1 + 4 + \left(\frac{2}{x}\right) + 4x \\
 &= 4x^2 + \left(\frac{1}{x^2}\right) + 5 + \left(\frac{2}{x}\right) + 4x
 \end{aligned}$$

Hence, $\left(2x + \frac{1}{x} + 1\right)^2 = 4x^2 + \left(\frac{1}{x^2}\right) + 5 + \left(\frac{2}{x}\right) + 4x$

Question 8(vi)

Find the square of:

$$5 - x + \frac{2}{x}$$

Answer

$$\left(5 - x + \frac{2}{x}\right)^2$$

Using the formula

$$\begin{aligned} (\because (x - y + z)^2 &= x^2 + y^2 + z^2 - 2xy - 2yz + 2xz) \\ &= (5)^2 + (x)^2 + \left(\frac{2}{x}\right)^2 - 2 \times 5 \times x - 2 \times x \times \left(\frac{2}{x}\right) + 2 \times \left(\frac{2}{x}\right) \times 5 \\ &= 25 + x^2 + \left(\frac{4}{x^2}\right) - 10x - \left(\frac{4x}{x}\right) + \left(\frac{20}{x}\right) \\ &= 25 + x^2 + \left(\frac{4}{x^2}\right) - 10x - 4 + \left(\frac{20}{x}\right) \\ &= (25 - 4) + x^2 + \left(\frac{4}{x^2}\right) - 10x + \left(\frac{20}{x}\right) \\ &= 21 + x^2 + \left(\frac{4}{x^2}\right) - 10x + \left(\frac{20}{x}\right) \end{aligned}$$

Hence, $\left(5 - x + \frac{2}{x}\right)^2 = 21 + x^2 + \left(\frac{4}{x^2}\right) - 10x + \left(\frac{20}{x}\right)$

Question 8(vii)

Find the square of:

$$2x - 3y + z$$

Answer

$$(2x - 3y + z)^2$$

Using the formula

$$(\because (x - y + z)^2 = x^2 + y^2 + z^2 - 2xy - 2yz + 2xz)$$

$$= (2x)^2 + (3y)^2 + (z)^2 - 2 \times (2x) \times (3y) - 2 \times (3y) \times (z) + 2 \times (z) \times (2x)$$

$$= 4x^2 + 9y^2 + z^2 - 12xy - 6yz + 4xz$$

$$\text{Hence, } (2x - 3y + z)^2 = 4x^2 + 9y^2 + z^2 - 12xy - 6yz + 4xz$$

Question 8(viii)

Find the square of:

$$x + \frac{1}{x} - 1$$

Answer

$$\left(x + \frac{1}{x} - 1\right)^2$$

Using the formula

$$(\because (x + y - z)^2 = x^2 + y^2 + z^2 + 2xy - 2yz - 2xz)$$

$$= (x)^2 + \left(\frac{1}{x}\right)^2 + (1)^2 + 2 \times x \times \left(\frac{1}{x}\right) - 2 \times \left(\frac{1}{x}\right) \times 1 - 2 \times 1 \times x$$

$$= x^2 + \left(\frac{1}{x^2}\right) + 1 + \left(\frac{2x}{x}\right) - \left(\frac{2}{x}\right) - 2x$$

$$= x^2 + \left(\frac{1}{x^2}\right) + 1 + 2 - \left(\frac{2}{x}\right) - 2x$$

$$= x^2 + \left(\frac{1}{x^2}\right) + 3 - \left(\frac{2}{x}\right) - 2x$$

Hence, $\left(x + \frac{1}{x} - 1\right)^2 = x^2 + \left(\frac{1}{x^2}\right) + 3 - \left(\frac{2}{x}\right) - 2x$

Question 9

Evaluate using expansion of $(a + b)^2$ or $(a - b)^2$:

(i) $(208)^2$

(ii) $(92)^2$

(iii) $(9.4)^2$

(iv) $(20.7)^2$

Answer

(i) $(208)^2$

$(200 + 8)^2$

Using the formula

$$\begin{aligned}(\because (x + y)^2 &= x^2 + 2xy + y^2) \\&= (200)^2 + 2 \times 200 \times 8 + (8)^2 \\&= 40000 + 3200 + 64 \\&= 43264\end{aligned}$$

$$\text{Hence, } (208)^2 = 43264$$

$$(ii) (92)^2$$

$$(100 - 8)^2$$

Using the formula

$$\begin{aligned}(\because (x - y)^2 &= x^2 - 2xy + y^2) \\&= (100)^2 - 2 \times 100 \times 8 + (8)^2 \\&= 10000 - 1600 + 64 \\&= 8400 + 64 \\&= 8464\end{aligned}$$

$$\text{Hence, } (92)^2 = 8464$$

$$(iii) (9.4)^2$$

$$(10 - 0.6)^2$$

Using the formula

$$\begin{aligned}(\because (x - y)^2 &= x^2 - 2xy + y^2) \\&= (10)^2 - 2 \times 10 \times 0.6 + (0.6)^2 \\&= 100 - 12 + 0.36 \\&= 88 + 0.36 \\&= 88.36\end{aligned}$$

Hence, $(9.4)^2 = 88.36$

$$\text{(iv) } (20.7)^2$$

$$(20 + 0.7)^2$$

Using the formula

$$\begin{aligned}(\because (x + y)^2 &= x^2 + 2xy + y^2) \\&= (20)^2 + 2 \times 20 \times 0.7 + (0.7)^2 \\&= 400 + 28 + 0.49 \\&= 428 + 0.49 \\&= 428.49\end{aligned}$$

Hence, $(20.7)^2 = 428.49$

Question 10(i)

Expand:

$$(2a + b)^3$$

Answer

$$(2a + b)^3$$

Using the formula,

$$(x + y)^3 = x^3 + y^3 + 3x^2y + 3xy^2$$

$$= (2a)^3 + b^3 + 3 \times (2a)^2 \times (b) + 3 \times (2a) \times (b)^2$$

$$= 8a^3 + b^3 + 12a^2b + 6ab^2$$

$$\text{Hence, } (2a + b)^3 = 8a^3 + b^3 + 12a^2b + 6ab^2$$

Question 10(ii)

Expand:

$$(a - 2b)^3$$

Answer

$$(a - 2b)^3$$

Using the formula,

$$(x - y)^3 = x^3 - y^3 - 3x^2y + 3xy^2$$

$$= a^3 - (2b)^3 - 3 \times (a)^2 \times (2b) + 3 \times (a) \times (2b)^2$$

$$= a^3 - 8b^3 - 6a^2b + 12ab^2$$

$$\text{Hence, } (a - 2b)^3 = a^3 - 8b^3 - 6a^2b + 12ab^2$$

Question 10(iii)

Expand:

$$(3x - 2y)^3$$

Answer

$$(3x - 2y)^3$$

Using the formula,

$$(x - y)^3 = x^3 - y^3 - 3x^2y + 3xy^2$$

$$= (3x)^3 - (2y)^3 - 3 \times (3x)^2 \times (2y) + 3 \times (3x) \times (2y)^2$$

$$= 27x^3 - 8y^3 - 54x^2y + 36xy^2$$

$$\text{Hence, } (3x - 2y)^3 = 27x^3 - 8y^3 - 54x^2y + 36xy^2$$

Question 10(iv)

Expand:

$$(x + 5y)^3$$

Answer

$$(x + 5y)^3$$

Using the formula,

$$\begin{aligned}(x + y)^3 &= x^3 + y^3 + 3x^2y + 3xy^2 \\&= (x)^3 + (5y)^3 + 3 \times (x)^2 \times (5y) + 3 \times (x) \times (5y)^2 \\&= x^3 + 125y^3 + 15x^2y + 75xy^2\end{aligned}$$

$$\text{Hence, } (x + 5y)^3 = x^3 + 125y^3 + 15x^2y + 75xy^2$$

Question 10(v)

Expand:

$$\left(a + \frac{1}{a}\right)^3$$

Answer

$$\left(a + \frac{1}{a}\right)^3$$

Using the formula,

$$\begin{aligned}(x + y)^3 &= x^3 + y^3 + 3x^2y + 3xy^2 \\&= (a)^3 + \left(\frac{1}{a}\right)^3 + 3 \times (a)^2 \times \left(\frac{1}{a}\right) + 3 \times (a) \times \left(\frac{1}{a}\right)^2 \\&= a^3 + \left(\frac{1}{a^3}\right) + \left(\frac{3a^2}{a}\right) + \left(\frac{3a}{a^2}\right) \\&= a^3 + \left(\frac{1}{a^3}\right) + 3a + \left(\frac{3}{a}\right)\end{aligned}$$

$$\text{Hence, } \left(a + \frac{1}{a}\right)^3 = a^3 + \left(\frac{1}{a^3}\right) + 3a + \left(\frac{3}{a}\right)$$

Question 10(vi)

Expand:

$$\left(2a - \frac{1}{2a}\right)^3$$

Answer

$$\left(2a - \frac{1}{2a}\right)^3$$

Using the formula,

$$(x - y)^3 = x^3 - y^3 - 3x^2y + 3xy^2$$

$$= (2a)^3 - \left(\frac{1}{2a}\right)^3 - 3 \times (2a)^2 \times \left(\frac{1}{2a}\right) + 3 \times (2a) \times \left(\frac{1}{2a}\right)^2$$

$$= 8a^3 - \left(\frac{1}{8a^3}\right) - \left(\frac{12a^2}{2a}\right) + \left(\frac{6a}{4a^2}\right)$$

$$= 8a^3 - \left(\frac{1}{8a^3}\right) - 6a + \left(\frac{3}{2a}\right)$$

$$\text{Hence, } \left(2a - \frac{1}{2a}\right)^3 = 8a^3 - \left(\frac{1}{8a^3}\right) - 6a + \left(\frac{3}{2a}\right)$$

Question 11(i)

Find the cube of:

$$a + 2$$

Answer

$$(a + 2)^3$$

Using the formula,

$$(x + y)^3 = x^3 + y^3 + 3x^2y + 3xy^2$$

$$= (a)^3 + (2)^3 + 3 \times (a)^2 \times (2) + 3 \times (a) \times (2)^2$$

$$= a^3 + 8 + 6a^2 + 12a$$

$$\text{Hence, } (a + 2)^3 = a^3 + 8 + 6a^2 + 12a$$

Question 11(ii)

Find the cube of:

$$2a - 1$$

Answer

$$(2a - 1)^3$$

Using the formula,

$$(x - y)^3 = x^3 - y^3 - 3x^2y + 3xy^2$$

$$= (2a)^3 - (1)^3 - 3 \times (2a)^2 \times (1) + 3 \times (2a) \times (1)^2$$

$$= 8a^3 - 1 - 12a^2 + 6a$$

$$\text{Hence, } (2a - 1)^3 = 8a^3 - 1 - 12a^2 + 6a$$

Question 11(iii)

Find the cube of:

$$2a + 3b$$

Answer

$$(2a + 3b)^3$$

Using the formula,

$$(x + y)^3 = x^3 + y^3 + 3x^2y + 3xy^2$$

$$= (2a)^3 + (3b)^3 + 3 \times (2a)^2 \times (3b) + 3 \times (2a) \times (3b)^2$$

$$= 8a^3 + 27b^3 + 36a^2b + 54ab^2$$

$$\text{Hence, } (2a + 3b)^3 = 8a^3 + 27b^3 + 36a^2b + 54ab^2$$

Question 11(iv)

Find the cube of:

$$3b - 2a$$

Answer

$$(3b - 2a)^3$$

Using the formula,

$$(x - y)^3 = x^3 - y^3 - 3x^2y + 3xy^2$$

$$= (3b)^3 - (2a)^3 + 3 \times (3b)^2 \times (2a) + 3 \times (3b) \times (2a)^2$$

$$= 27b^3 - 8a^3 - 54b^2a + 36ba^2$$

$$\text{Hence, } (3b - 2a)^3 = 27b^3 - 8a^3 - 54b^2a + 36ba^2$$

Question 11(v)

Find the cube of:

$$2x + \frac{1}{x}$$

Answer

$$2x + \frac{1}{x}$$

Using the formula,

$$(x + y)^3 = x^3 + y^3 + 3x^2y + 3xy^2$$

$$= (2x)^3 + \left(\frac{1}{x}\right)^3 + 3 \times (2x)^2 \times \left(\frac{1}{x}\right) + 3 \times 2x \times \left(\frac{1}{x}\right)^2$$

$$= 8x^3 + \left(\frac{1}{x^3}\right) + \left(\frac{12x^2}{x}\right) + \left(\frac{6x}{x^2}\right)$$

$$= 8x^3 + \left(\frac{1}{x^3}\right) + 12x + \frac{6}{x}$$

$$\text{Hence, } 2x + \frac{1}{x} = 8x^3 + \left(\frac{1}{x^3}\right) + 12x + \frac{6}{x}$$

Question 11(vi)

Find the cube of:

$$x - \frac{1}{2}$$

Answer

$$x - \frac{1}{2}$$

Using the formula,

$$(x - y)^3 = x^3 - y^3 - 3x^2y + 3xy^2$$

$$= (x)^3 - \left(\frac{1}{2}\right)^3 - 3 \times x^2 \times \left(\frac{1}{2}\right) + 3 \times x \times \left(\frac{1}{2}\right)^2$$

$$= x^3 - \left(\frac{1}{8}\right) - \left(\frac{3x^2}{2}\right) + \left(\frac{3x}{4}\right)$$

$$\text{Hence, } x + \frac{1}{2} = x^3 - \left(\frac{1}{8}\right) - \left(\frac{3x^2}{2}\right) + \left(\frac{3x}{4}\right)$$

Exercise 12(B)

Question 1(i)

If $x - \frac{1}{x} = 3$, the value of $x^2 + \frac{1}{x^2}$ is:

1. 0

2. 7

3. 11

4. 9

Answer

Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$\left(x - \frac{1}{x}\right)^2 = x^2 - 2 \times x \times \frac{1}{x} + \left(\frac{1}{x}\right)^2$$

$$= x^2 - \frac{2x}{x} + \left(\frac{1}{x}\right)^2$$

$$= x^2 - 2 + \frac{1}{x^2}$$

$$\text{Putting the value, } x - \frac{1}{x} = 3$$

$$(3)^2 = x^2 - 2 + \frac{1}{x^2}$$

$$\Rightarrow 9 = x^2 - 2 + \frac{1}{x^2}$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 9 + 2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 11$$

Hence, option 3 is the correct option.

Question 1(ii)

If $a + b = 7$ and $ab = 10$; the value of $a^2 + b^2$ is equal to:

1. 29

2. 49

3. 39

4. 69

Answer

Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$(a + b)^2 = a^2 + 2ab + b^2$$

Putting the value, $a + b = 7$ and $ab = 10$

$$\Rightarrow (7)^2 = a^2 + 2 \times 10 + b^2$$

$$\Rightarrow 49 = a^2 + 20 + b^2$$

$$\Rightarrow a^2 + b^2 = 49 - 20$$

$$\Rightarrow a^2 + b^2 = 29$$

Hence, option 1 is the correct option.

Question 1(iii)

The value of $\frac{95 \times 95 - 5 \times 5}{95 - 5}$ is equal to:

1. 100
2. 90
3. 95
4. none of these

Answer

Using the formula,

$$[\because (x^2 - y^2) = (x + y)(x - y)]$$

$$\frac{95 \times 95 - 5 \times 5}{95 - 5}$$

$$\frac{(95)^2 - (5)^2}{95 - 5}$$

$$= \frac{(95 - 5)(95 + 5)}{95 - 5}$$

$$= \frac{\cancel{(95 - 5)}(95 + 5)}{\cancel{(95 - 5)}}$$

$$= 95 + 5$$

$$= 100$$

Hence, option 1 is the correct option.

Question 1(iv)

If $a - b = 1$ and $a + b = 3$, the value of ab is:

1. 4

2. 2

3. -2

4. 0

Answer

Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

And

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

Putting $a + b = 3$, we get

$$(3)^2 = a^2 + 2ab + b^2$$

$$9 = a^2 + 2ab + b^2$$

$$9 - 2ab = a^2 + b^2 \dots\dots\dots(1)$$

And,

$$(a - b)^2 = a^2 - 2ab + b^2$$

Putting the value, $a - b = 1$

$$(1)^2 = a^2 - 2ab + b^2$$

$$1 = a^2 + b^2 - 2ab$$

Using equation (1)

$$1 = 9 - 2ab - 2ab$$

$$1 = 9 - 4ab$$

$$4ab = 9 - 1$$

$$4ab = 8$$

$$ab = \frac{8}{4}$$

$$ab = 2$$

Hence, option 2 is the correct option.

Question 1(v)

If $x + \frac{1}{x} = 2$, the value of $\left(x^3 + \frac{1}{x^3}\right) - \left(x^2 + \frac{1}{x^2}\right)$ is:

1. 0

2. 4

3. 2

4. 6

Answer

Using the formula,

$$[\because (x + y)^3 = x^3 + 3xy(x + y) + y^3]$$

And,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

$$\left(x + \frac{1}{x}\right)^3 = x^3 + 3 \times x \times \frac{1}{x} \left(x + \frac{1}{x}\right) + \left(\frac{1}{x}\right)^3$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^3 = x^3 + 3\left(x + \frac{1}{x}\right) + \frac{1}{x^3}$$

$$\text{Putting } x + \frac{1}{x} = 2$$

$$2^3 = x^3 + 3 \times 2 + \frac{1}{x^3}$$

$$\Rightarrow 8 = x^3 + 6 + \frac{1}{x^3}$$

$$\Rightarrow x^3 + \frac{1}{x^3} = 8 - 6$$

$$\Rightarrow x^3 + \frac{1}{x^3} = 2 \dots \dots \dots (1)$$

And ,

$$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 \times x \times \frac{1}{x} + \left(\frac{1}{x}\right)^2$$

$$\Rightarrow \left(x + \frac{1}{x}\right)^2 = x^2 + 2 + \frac{1}{x^2}$$

Putting the value $x + \frac{1}{x} = 2$, we get

$$2^2 = x^2 + 2 + \frac{1}{x^2}$$

$$\Rightarrow 4 = x^2 + 2 + \frac{1}{x^2}$$

$$\Rightarrow 4 - 2 = x^2 + \frac{1}{x^2}$$

$$\Rightarrow 2 = x^2 + \frac{1}{x^2} \dots \dots \dots (2)$$

Now, using equation (1) and (2),

$$\left(x^3 + \frac{1}{x^3}\right) - \left(x^2 + \frac{1}{x^2}\right)$$

$$= 2 - 2$$

$$= 0$$

Hence, option 1 is the correct option.

Question 2

If $a + b = 5$ and $ab = 6$, find $a^2 + b^2$

Answer

Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$(a + b)^2 = a^2 + 2ab + b^2$$

Putting the value, $a + b = 5$ and $ab = 6$

$$\Rightarrow (5)^2 = a^2 + 2 \times 6 + b^2$$

$$\Rightarrow 25 = a^2 + 12 + b^2$$

$$\Rightarrow a^2 + b^2 = 25 - 12$$

$$\Rightarrow a^2 + b^2 = 13$$

Hence, the value of $(a^2 + b^2)$ is 13.

Question 3

If $a - b = 6$ and $ab = 16$, find $a^2 + b^2$

Answer

Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

So,

$$(a - b)^2 = a^2 - 2ab + b^2$$

Putting the value, $a - b = 6$ and $ab = 16$

$$\Rightarrow (6)^2 = a^2 - 2 \times 16 + b^2$$

$$\Rightarrow 36 = a^2 - 32 + b^2$$

$$\Rightarrow a^2 + b^2 = 36 + 32$$

$$\Rightarrow a^2 + b^2 = 68$$

Hence, the value of $(a^2 + b^2)$ is 68.

Question 4

If $a^2 + b^2 = 29$ and $ab = 10$, find:

(i) $a + b$

(ii) $a - b$

Answer

(i) Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$(a + b)^2 = a^2 + 2ab + b^2$$

Putting the value, $a^2 + b^2 = 29$ and $ab = 10$

$$\Rightarrow (a + b)^2 = (a^2 + b^2) + 2ab$$

$$\Rightarrow (a + b)^2 = (29) + 2 \times 10$$

$$\Rightarrow (a + b)^2 = 29 + 20$$

$$\Rightarrow (a + b)^2 = 49$$

$$\Rightarrow a + b = \sqrt{49}$$

$$\Rightarrow a + b = 7 \text{ or } -7$$

Hence, the values of $(a + b)$ are 7 or -7.

(ii) Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

So,

$$(a - b)^2 = a^2 - 2ab + b^2$$

Putting the value, $a^2 + b^2 = 29$ and $ab = 10$

$$\Rightarrow (a - b)^2 = (a^2 + b^2) - 2ab$$

$$\Rightarrow (a - b)^2 = (29) - 2 \times 10$$

$$\Rightarrow (a - b)^2 = 29 - 20$$

$$\Rightarrow (a - b)^2 = 9$$

$$\Rightarrow a - b = \sqrt{9}$$

$$\Rightarrow a - b = 3 \text{ or } -3$$

Hence, the values of $(a - b)$ are 3 or -3.

Question 5

If $a^2 + b^2 = 10$ and $ab = 3$, find:

(i) $a - b$

(ii) $a + b$

Answer

(i) Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

So,

$$(a - b)^2 = a^2 - 2ab + b^2$$

Putting the value, $a^2 + b^2 = 10$ and $ab = 3$

$$\Rightarrow (a - b)^2 = (a^2 + b^2) - 2ab$$

$$\Rightarrow (a - b)^2 = (10) - 2 \times 3$$

$$\Rightarrow (a - b)^2 = 10 - 6$$

$$\Rightarrow (a - b)^2 = 4$$

$$\Rightarrow a - b = \sqrt{4}$$

$$\Rightarrow a - b = 2 \text{ or } -2$$

Hence, the values of $(a - b)$ are 2 or -2.

(ii) Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$(a + b)^2 = a^2 + 2ab + b^2$$

Putting the value, $a^2 + b^2 = 10$ and $ab = 3$

$$\Rightarrow (a + b)^2 = (a^2 + b^2) + 2ab$$

$$\Rightarrow (a + b)^2 = (10) + 2 \times 3$$

$$\Rightarrow (a + b)^2 = 10 + 6$$

$$\Rightarrow (a + b)^2 = 16$$

$$\Rightarrow a + b = \sqrt{16}$$

$$\Rightarrow a + b = 4 \text{ or } -4$$

Hence, the values of $(a + b)$ are 4 or -4.

Question 6

If $a + \frac{1}{a} = 3$, find: $a^2 + \frac{1}{a^2}$

Answer

Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$\left(a + \frac{1}{a}\right)^2 = a^2 + 2 \times a \times \frac{1}{a} + \left(\frac{1}{a}\right)^2$$

$$\Rightarrow \left(a + \frac{1}{a}\right)^2 = a^2 + 2 + \frac{1}{a^2}$$

Putting the value $a + \frac{1}{a} = 3$, we get

$$3^2 = a^2 + 2 + \frac{1}{a^2}$$

$$\Rightarrow 9 = a^2 + 2 + \frac{1}{a^2}$$

$$\Rightarrow a^2 + \frac{1}{a^2} = 9 - 2$$

$$\Rightarrow a^2 + \frac{1}{a^2} = 7$$

Hence, the value of $a^2 + \frac{1}{a^2}$ is 7.

Question 7

If $a - \frac{1}{a} = 4$, find: $a^2 + \frac{1}{a^2}$

Answer

Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

So,

$$\left(a - \frac{1}{a}\right)^2 = a^2 - 2 \times a \times \frac{1}{a} + \left(\frac{1}{a}\right)^2$$

$$\Rightarrow \left(a - \frac{1}{a}\right)^2 = a^2 - 2 + \frac{1}{a^2}$$

Putting the value $a - \frac{1}{a} = 4$, we get

$$4^2 = a^2 - 2 + \frac{1}{a^2}$$

$$\Rightarrow 16 = a^2 - 2 + \frac{1}{a^2}$$

$$\Rightarrow a^2 + \frac{1}{a^2} = 16 + 2$$

$$\Rightarrow a^2 + \frac{1}{a^2} = 18$$

Hence, the value of $a^2 + \frac{1}{a^2}$ is 18.

Question 8

If $a^2 + \frac{1}{a^2} = 23$, find: $a + \frac{1}{a}$

Answer

Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$\left(a + \frac{1}{a}\right)^2 = a^2 + 2 \times a \times \frac{1}{a} + \left(\frac{1}{a}\right)^2$$

$$\Rightarrow \left(a + \frac{1}{a}\right)^2 = a^2 + 2 + \frac{1}{a^2}$$

Putting the value $a^2 + \frac{1}{a^2} = 23$, we get

$$\Rightarrow \left(a + \frac{1}{a}\right)^2 = \left(a^2 + \frac{1}{a^2}\right) + 2$$

$$\Rightarrow \left(a + \frac{1}{a}\right)^2 = 23 + 2$$

$$\Rightarrow \left(a + \frac{1}{a}\right)^2 = 25$$

$$\Rightarrow \left(a + \frac{1}{a}\right) = \sqrt{25}$$

$$\Rightarrow \left(a + \frac{1}{a}\right) = 5 \text{ or } -5$$

Hence, the values of $\left(a + \frac{1}{a}\right)$ are 5 or -5.

Question 9

If $a^2 + \frac{1}{a^2} = 11$, find: $a - \frac{1}{a}$

Answer

Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

So,

$$\left(a - \frac{1}{a}\right)^2 = a^2 - 2 \times a \times \frac{1}{a} + \left(\frac{1}{a}\right)^2$$

$$\Rightarrow \left(a - \frac{1}{a}\right)^2 = a^2 - 2 + \frac{1}{a^2}$$

Putting the value $a^2 + \frac{1}{a^2} = 11$, we get

$$\Rightarrow \left(a - \frac{1}{a}\right)^2 = \left(a^2 + \frac{1}{a^2}\right) - 2$$

$$\Rightarrow \left(a - \frac{1}{a}\right)^2 = 11 - 2$$

$$\Rightarrow \left(a - \frac{1}{a}\right)^2 = 9$$

$$\Rightarrow \left(a - \frac{1}{a}\right) = \sqrt{9}$$

$$\Rightarrow \left(a - \frac{1}{a}\right) = 3 \text{ or } -3$$

Hence, the values of $\left(a - \frac{1}{a}\right)$ are 3 or -3.

Question 10

$a + b + c = 10$ and $a^2 + b^2 + c^2 = 38$, find $ab + bc + ca$

Answer

Using the formula,

$$[\because (x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx]$$

So,

$$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

Putting $(a + b + c) = 10$ and $(a^2 + b^2 + c^2) = 38$, we get

$$\Rightarrow (10)^2 = 38 + 2(ab + bc + ca)$$

$$\Rightarrow 100 = 38 + 2(ab + bc + ca)$$

$$\Rightarrow 2(ab + bc + ca) = 100 - 38$$

$$\Rightarrow 2(ab + bc + ca) = 62$$

$$\Rightarrow ab + bc + ca = \frac{62}{2}$$

$$\Rightarrow ab + bc + ca = 31$$

Hence, the value of $(ab + bc + ca)$ is 31.

Question 11

Find : $a^2 + b^2 + c^2$, if $a + b + c = 9$ and $ab + bc + ca = 24$.

Answer

Using the formula,

$$[\because (x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx]$$

So,

$$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

Putting $(a + b + c) = 9$ and $(ab + bc + ca) = 24$, we get

$$\Rightarrow (9)^2 = a^2 + b^2 + c^2 + 2 \times 24$$

$$\Rightarrow 81 = a^2 + b^2 + c^2 + 48$$

$$\Rightarrow a^2 + b^2 + c^2 = 81 - 48$$

$$\Rightarrow a^2 + b^2 + c^2 = 33$$

Hence, the value of $a^2 + b^2 + c^2$ is 33.

Question 12

Find: $a + b + c$, if $a^2 + b^2 + c^2 = 83$ and $ab + bc + ca = 71$

Answer

Using the formula,

$$[\because (x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx]$$

So,

$$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

Putting $(a^2 + b^2 + c^2) = 83$ and $(ab + bc + ca) = 71$, we get

$$\Rightarrow (a + b + c)^2 = 83 + 2 \times 71$$

$$\Rightarrow (a + b + c)^2 = 83 + 142$$

$$\Rightarrow (a + b + c)^2 = 225$$

$$\Rightarrow a + b + c = \sqrt{225}$$

$$\Rightarrow a + b + c = 15 \text{ or } -15$$

Hence, the values of $(a + b + c)$ are 15 or -15.

Question 13

If $a + b = 6$ and $ab = 8$, find: $a^3 + b^3$.

Answer

Using the formula,

$$[\because (x + y)^3 = x^3 + y^3 + 3xy(x + y)]$$

So,

$$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

Putting the values $a + b = 6$ and $ab = 8$, we get

$$\Rightarrow (6)^3 = a^3 + b^3 + 3 \times 8 \times 6$$

$$\Rightarrow 216 = a^3 + b^3 + 144$$

$$\Rightarrow a^3 + b^3 = 216 - 144$$

$$\Rightarrow a^3 + b^3 = 72$$

Hence, the value of $a^3 + b^3$ is 72.

Question 14

If $a - b = 3$ and $ab = 10$, find: $a^3 - b^3$.

Answer

Using the formula,

$$[\because (x - y)^3 = x^3 - y^3 - 3xy(x - y)]$$

So,

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

Putting the values $a - b = 3$ and $ab = 10$, we get

$$\Rightarrow (3)^3 = a^3 - b^3 - 3 \times 10 \times 3$$

$$\Rightarrow 27 = a^3 - b^3 - 90$$

$$\Rightarrow a^3 + b^3 = 27 + 90$$

$$\Rightarrow a^3 + b^3 = 117$$

Hence, the value of $a^3 + b^3$ is 117.

Question 15

Find : $a^3 + \frac{1}{a^3}$, if $a + \frac{1}{a} = 5$.

Answer

Using the formula,

$$[\because (x + y)^3 = x^3 + 3xy(x + y) + y^3]$$

So,

$$\left(a + \frac{1}{a}\right)^3 = a^3 + 3 \times a \times \frac{1}{a} \left(a + \frac{1}{a}\right) + \left(\frac{1}{a}\right)^3$$

$$\Rightarrow \left(a + \frac{1}{a}\right)^3 = a^3 + 3\left(a + \frac{1}{a}\right) + \frac{1}{a^3}$$

$$\text{Putting } a + \frac{1}{a} = 5$$

$$5^3 = a^3 + 3 \times 5 + \frac{1}{a^3}$$

$$\Rightarrow 125 = a^3 + 15 + \frac{1}{a^3}$$

$$\Rightarrow a^3 + \frac{1}{a^3} = 125 - 15$$

$$\Rightarrow a^3 + \frac{1}{a^3} = 110$$

Hence, the value of $a^3 + \frac{1}{a^3}$ is 110.

Question 16

Find : $a^3 - \frac{1}{a^3}$, if $a - \frac{1}{a} = 4$.

Answer

Using the formula,

$$[\because (x - y)^3 = x^3 - 3xy(x - y) - y^3]$$

So,

$$\left(a - \frac{1}{a}\right)^3 = a^3 - 3 \times a \times \frac{1}{a} \left(a - \frac{1}{a}\right) - \left(\frac{1}{a}\right)^3$$

$$\Rightarrow \left(a - \frac{1}{a}\right)^3 = a^3 - 3\left(a - \frac{1}{a}\right) - \frac{1}{a^3}$$

$$\text{Putting } a - \frac{1}{a} = 4$$

$$4^3 = a^3 - 3 \times 4 - \frac{1}{a^3}$$

$$\Rightarrow 64 = a^3 - 12 - \frac{1}{a^3}$$

$$\Rightarrow a^3 - \frac{1}{a^3} = 64 + 12$$

$$\Rightarrow a^3 - \frac{1}{a^3} = 76$$

Hence, the value of $a^3 - \frac{1}{a^3}$ is 76.

Question 17

If $2x - \frac{1}{2x} = 4$, find:

(i) $4x^2 + \frac{1}{4x^2}$

(ii) $8x^3 - \frac{1}{8x^3}$

Answer

(i) Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

So,

$$\left(2x - \frac{1}{2x}\right)^2 = (2x)^2 - 2 \times 2x \times \frac{1}{2x} + \left(\frac{1}{2x}\right)^2$$

$$\Rightarrow \left(2x - \frac{1}{2x}\right)^2 = 4x^2 - 2 + \frac{1}{4x^2}$$

Putting the value $2x - \frac{1}{2x} = 4$, we get

$$4^2 = 4x^2 - 2 + \frac{1}{4x^2}$$

$$\Rightarrow 16 = 4x^2 - 2 + \frac{1}{4x^2}$$

$$\Rightarrow 4x^2 + \frac{1}{4x^2} = 16 + 2$$

$$\Rightarrow 4x^2 + \frac{1}{4x^2} = 18$$

Hence, the value of $4x^2 + \frac{1}{4x^2}$ is 18.

(ii) Using the formula,

$$[\because (x - y)^3 = x^3 - 3xy(x - y) - y^3]$$

So,

$$\left(2x - \frac{1}{2x}\right)^3 = (2x)^3 - 3 \times 2x \times \frac{1}{2x} \left(2x - \frac{1}{2x}\right) - \left(\frac{1}{2x}\right)^3$$

$$\Rightarrow \left(2x - \frac{1}{2x}\right)^3 = 8x^3 - 3\left(2x - \frac{1}{2x}\right) - \frac{1}{8x^3}$$

Putting $2x - \frac{1}{2x} = 4$

$$4^3 = 8x^3 - 3 \times 4 - \frac{1}{8x^3}$$

$$\Rightarrow 64 = 8x^3 - 12 - \frac{1}{8x^3}$$

$$\Rightarrow 8x^3 - \frac{1}{8x^3} = 64 + 12$$

$$\Rightarrow 8x^3 - \frac{1}{8x^3} = 76$$

Hence, the value of $8x^3 - \frac{1}{8x^3} = 76$.

Question 18

If $3x + \frac{1}{3x} = 3$, find:

(i) $9x^2 + \frac{1}{9x^2}$

(ii) $27x^3 + \frac{1}{27x^3}$

Answer

(i) Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$\left(3x + \frac{1}{3x}\right)^2 = (3x)^2 + 2 \times 3x \times \frac{1}{3x} + \left(\frac{1}{3x}\right)^2$$

$$\Rightarrow \left(3x + \frac{1}{3x}\right)^2 = 9x^2 + 2 + \frac{1}{9x^2}$$

Putting the value $3x + \frac{1}{3x} = 3$, we get

$$3^2 = 9x^2 + 2 + \frac{1}{9x^2}$$

$$\Rightarrow 9 = 9x^2 + 2 + \frac{1}{9x^2}$$

$$\Rightarrow 9x^2 + \frac{1}{9x^2} = 9 - 2$$

$$\Rightarrow 9x^2 + \frac{1}{9x^2} = 7$$

Hence, the value of $9x^2 + \frac{1}{9x^2}$ is 7.

(ii) Using the formula,

$$[\because (x + y)^3 = x^3 + 3xy(x + y) + y^3]$$

So,

$$\left(3x + \frac{1}{3x}\right)^3 = (3x)^3 + 3 \times 3x \times \frac{1}{3x} \left(3x + \frac{1}{3x}\right) + \left(\frac{1}{3x}\right)^3$$

$$\Rightarrow \left(3x + \frac{1}{3x}\right)^3 = 27x^3 + 3\left(3x + \frac{1}{3x}\right) + \frac{1}{27x^3}$$

$$\text{Putting } 3x + \frac{1}{3x} = 3$$

$$3^3 = 27x^3 + 3 \times 3 + \frac{1}{27x^3}$$

$$\Rightarrow 27 = 27x^3 + 9 + \frac{1}{27x^3}$$

$$\Rightarrow 27x^3 + \frac{1}{27x^3} = 27 - 9$$

$$\Rightarrow 27x^3 + \frac{1}{27x^3} = 18$$

Hence, the value of $27x^3 + \frac{1}{27x^3}$ is 18.

Question 19

The sum of the squares of two numbers is 13 and their product is 6. Find :

- (i) the sum of the two numbers.
- (ii) the difference between them.

Answer

(i) Let 2 numbers be x and y. The sum of the squares of two numbers is 13 and their product is 6. So,

$$x^2 + y^2 = 13$$

And,

$$xy = 6$$

Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

$$(x + y)^2 = (x^2 + y^2) + 2xy$$

Putting the value,

$$\Rightarrow (x + y)^2 = 13 + 2 \times 6$$

$$\Rightarrow (x + y)^2 = 13 + 12$$

$$\Rightarrow (x + y)^2 = 25$$

$$\Rightarrow x + y = \sqrt{25}$$

$$\Rightarrow x + y = 5 \text{ or } -5$$

Hence, the sum of the two numbers = 5 or -5.

(ii) Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$(x - y)^2 = (x^2 + y^2) - 2xy$$

Putting the value,

$$\Rightarrow (x - y)^2 = 13 - 2 \times 6$$

$$\Rightarrow (x - y)^2 = 13 - 12$$

$$\Rightarrow (x - y)^2 = 1$$

$$\Rightarrow x - y = \sqrt{1}$$

$$\Rightarrow x - y = 1 \text{ or } -1$$

Hence, the difference of the two numbers = 1 or -1.

Test Yourself

Question 1(i)

If a is positive and $a^2 + \frac{1}{a^2} = 18$; then the value of $a - \frac{1}{a}$ is:

1. 4
2. 16
3. 20
4. $2\sqrt{5}$

Answer

Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$\left(a - \frac{1}{a}\right)^2 = a^2 - 2 \times a \times \frac{1}{a} + \left(\frac{1}{a}\right)^2$$

$$= a^2 - \frac{2a}{a} + \left(\frac{1}{a}\right)^2$$

$$= a^2 - 2 + \left(\frac{1}{a}\right)^2$$

Putting the value, $a^2 + \frac{1}{a^2} = 18$

$$\Rightarrow \left(a - \frac{1}{a}\right)^2 = a^2 + \left(\frac{1}{a}\right)^2 - 2$$

$$\Rightarrow \left(a - \frac{1}{a}\right)^2 = 18 - 2$$

$$\Rightarrow \left(a - \frac{1}{a}\right)^2 = 16$$

$$\Rightarrow \left(a - \frac{1}{a}\right) = \sqrt{16}$$

$$\Rightarrow \left(a - \frac{1}{a}\right) = 4 \text{ or } -4$$

From options, only value matching is 4.

So, $\left(a - \frac{1}{a}\right) = 4$

Hence, option 1 is the correct option.

Question 1(ii)

$(x + y)(x - y)(x^2 + y^2)(x^4 + y^4)$ is equal to:

1. $x^4 + y^4$

2. $x^8 + y^8$

3. $x^8 - y^8$

4. $2x^6y^6$

Answer

Using the formula,

$$[\because (x - y)(x + y) = x^2 - y^2]$$

$$(x + y)(x - y)(x^2 + y^2)(x^4 + y^4)$$

$$\Rightarrow [(x + y)(x - y)](x^2 + y^2)(x^4 + y^4)$$

$$\Rightarrow [(x^2 - y^2)(x^2 + y^2)](x^4 + y^4)$$

$$\Rightarrow [(x^4 - y^4)(x^4 + y^4)]$$

$$\Rightarrow (x^8 - y^8)$$

Hence, option 3 is the correct option.

Question 1(iii)

The value of 102×98 is:

1. 6999

2. 6696

3. 9696

4. 9996

Answer

Using the formula,

$$[\because (x - y)(x + y) = x^2 - y^2]$$

$$102 \times 98 = (100 + 2)(100 - 2)$$

$$= 100^2 - 2^2$$

$$= 10000 - 4$$

$$= 9996$$

Hence, option 4 is the correct option.

Question 1(iv)

$(x + 3)(x + 3) - (x - 2)(x - 2)$ is equal to:

1. $10x + 5$
2. $10x - 5$
3. $5 - 10x$
4. none of these

Answer

Using the formula,

$$[\because (x + y)^2 = x^2 + y^2 + 2xy]$$

And,

$$[\because (x - y)^2 = x^2 + y^2 - 2xy]$$

$$(x + 3)(x + 3) - (x - 2)(x - 2)$$

$$\Rightarrow [(x + 3)^2] - [(x - 2)^2]$$

$$\Rightarrow [x^2 + 3^2 + 2 \times x \times 3] - [x^2 + 2^2 - 2 \times x \times 2]$$

$$\Rightarrow [x^2 + 9 + 6x] - [x^2 + 4 - 4x]$$

$$\Rightarrow x^2 + 9 + 6x - x^2 - 4 + 4x$$

$$\Rightarrow (x^2 - x^2) + (9 - 4) + (6x + 4x)$$

$$\Rightarrow 5 + 10x$$

$$\Rightarrow 10x + 5$$

Hence, option 1 is the correct option.

Question 1(v)

If $5a = 30^2 - 25^2$, the value of a is:

1. 5

2. 11

3. 55

4. none of these

Answer

Using the formula,

$$[\because (x - y)(x + y) = x^2 - y^2]$$

$$5a = 30^2 - 25^2$$

$$\Rightarrow 5a = (30 - 25) (30 + 25)$$

$$\Rightarrow 5a = 5 \times 55$$

$$\Rightarrow 5a = 275$$

$$\Rightarrow a = \frac{275}{5}$$

$$\Rightarrow a = 55$$

Hence, option 3 is the correct option.

Question 1(vi)

Statement 1: Cube of a binomial : $(a - b)^3 = a^3 + 3a^2b - 3ab^2 - b^3$

Statement 2: $(a - b)^2 - (a + b)^2 = 4ab$.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

Cube of a binomial:

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

$$= a^3 - b^3 - 3a^2b + 3ab^2$$

$$= a^3 - 3a^2b + 3ab^2 - b^3$$

So, statement 1 is false.

Solving,

$$(a - b)^2 - (a + b)^2$$

$$= [a^2 - 2ab + b^2] - [a^2 + 2ab + b^2]$$

$$= a^2 - 2ab + b^2 - a^2 - 2ab - b^2$$

$$= -4ab.$$

So, statement 2 is false.

Hence, Option 2 is the correct option.

Question 1(vii)

Assertion (A) : Using appropriate identity, we get $22.5 \times 21.5 = 484.75$.

Reason (R) : The product: $(x + y)(x - y) = x^2 - y^2$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.

3. A is true, but R is false.

4. A is false, but R is true.

Answer

Solving,

$$(x + y)(x - y) = x(x - y) + y(x - y)$$

$$= x^2 - xy + xy - y^2$$

$$= x^2 - y^2$$

So, reason (R) is true.

Solving,

$$\Rightarrow 22.5 \times 21.5$$

$$\Rightarrow (22 + 0.5)(22 - 0.5)$$

Using the identity; $(x + y)(x - y) = x^2 - y^2$

$$= (22)^2 - (0.5)^2$$

$$= 484 - 0.25$$

$$= 483.75$$

So, assertion (A) is false.

Hence, option 4 is the correct option.

Question 1(viii)

Assertion (A) : If we add 9 with $49x^2 - 42x$, the resultant expression will be a perfect square expression.

Reason (R) : The product of the sum and difference of the same two terms = Difference of their squares.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given: $49x^2 - 42x$

Adding 9 in it the term the expression becomes

$$\Rightarrow 49x^2 - 42x + 9$$

$$\Rightarrow (7x)^2 - 2 \times (7x) \times 3 + (3)^2$$

$$\Rightarrow (7x - 3)^2$$

The resultant expression will be a perfect square expression.

So, assertion (A) is true.

The product of the sum and difference of the same two terms = Difference of their squares.

$$(a + b)(a - b) = a^2 - b^2$$

So, reason (R) is true but reason (R) is not the correct explanation of assertion (A).

Hence, option 2 is the correct option.

Question 1(ix)

Assertion (A) : If the volume of a cube is $a^3 + b^3 + 3ab(a + b)$, then the edge of the cube is $(a + b)$

Reason (R) : $(1^{\text{st}} \text{ term} + 2^{\text{nd}} \text{ term})^3 = (1^{\text{st}} \text{ term})^3 + 3(1^{\text{st}} \text{ term})^2 \cdot (2^{\text{nd}} \text{ term}) + 3(2^{\text{nd}} \text{ term})^2 \cdot (1^{\text{st}} \text{ term}) + (2^{\text{nd}} \text{ term})^3$.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

$$\text{Volume of a cube} = (\text{edge})^3$$

$$\Rightarrow (\text{edge})^3 = a^3 + b^3 + 3ab(a + b)$$

$$\Rightarrow (\text{edge})^3 = (a + b)^3$$

$$\Rightarrow \text{edge} = \sqrt[3]{(a + b)^3}$$

$$\Rightarrow \text{edge} = (a + b)$$

So, assertion (A) is true.

Using identity,

$$\Rightarrow (a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$\Rightarrow (a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

Substitute $a = 1^{\text{st}}$ term and $b = 2^{\text{nd}}$ term

$$(1^{\text{st}} \text{ term} + 2^{\text{nd}} \text{ term})^3 = (1^{\text{st}} \text{ term})^3 + 3(1^{\text{st}} \text{ term})^2 \cdot (2^{\text{nd}} \text{ term}) + 3(2^{\text{nd}} \text{ term})^2 \cdot (1^{\text{st}} \text{ term}) + (2^{\text{nd}} \text{ term})^3$$

So, reason (R) is true and, reason (R) is the correct explanation of assertion (A).

Hence, option 1 is the correct option.

Question 1(x)

Assertion (A) : $687 \times 687 - 313 \times 313 = 37400$

Reason (R) : The product of the sum and difference of the same two terms = Difference of their squares.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

We know that,

The product of the sum and difference of the same two terms = Difference of their squares.

$$(a + b)(a - b) = a^2 - b^2$$

So, reason (R) is true.

Given,

$$\Rightarrow 687 \times 687 - 313 \times 313 = 34700$$

Solving, L.H.S.

$$\Rightarrow 687 \times 687 - 313 \times 313$$

$$\Rightarrow 687^2 - 313^2$$

$$\Rightarrow (687 - 313)(687 + 313)$$

$$\Rightarrow 374 \times 1000$$

$$\Rightarrow 374000.$$

Since, L.H.S. $\neq$ R.H.S.

So, assertion (A) is false.

Hence, option 4 is the correct option.

Question 2(i)

Evaluate :

$$\left(3x + \frac{1}{2}\right) \left(2x + \frac{1}{3}\right)$$

Answer

$$\left(3x + \frac{1}{2}\right) \left(2x + \frac{1}{3}\right)$$

$$= 3x \left(2x + \frac{1}{3}\right) + \frac{1}{2} \left(2x + \frac{1}{3}\right)$$

$$= 3x \times 2x + 3x \times \frac{1}{3} + \frac{1}{2} \times 2x + \frac{1}{2} \times \frac{1}{3}$$

$$= 6x^2 + \frac{3x}{3} + \frac{2x}{2} + \frac{1 \times 1}{2 \times 3}$$

$$= 6x^2 + x + x + \frac{1}{6}$$

$$= 6x^2 + 2x + \frac{1}{6}$$

$$\text{Hence, } \left(3x + \frac{1}{2}\right) \left(2x + \frac{1}{3}\right) = 6x^2 + 2x + \frac{1}{6}$$

Question 2(ii)

Evaluate :

$$(2a + 0.5)(7a - 0.3)$$

Answer

$$(2a + 0.5)(7a - 0.3)$$

$$= 2a (7a - 0.3) + 0.5 (7a - 0.3)$$

$$= 14a^{(1+1)} - 0.6a + 3.5a - 0.15$$

$$= 14a^2 + 2.9a - 0.15$$

$$\text{Hence, } (2a + 0.5)(7a - 0.3) = 14a^2 + 2.9a - 0.15$$

Question 2(iii)

Evaluate :

$$(9 - y) (7 + y)$$

Answer

$$(9 - y) (7 + y)$$

$$= 9 (7 + y) - y (7 + y)$$

$$= 63 + 9y - 7y - y^{(1+1)}$$

$$= 63 + 2y - y^2$$

$$\text{Hence, } (9 - y) (7 + y) = 63 + 2y - y^2$$

Question 2(iv)

Evaluate :

$$(2 - z) (15 - z)$$

Answer

$$\begin{aligned}(2 - z) (15 - z) \\&= 2 (15 - z) - z (15 - z) \\&= 30 - 2z - 15z + z^{(1+1)} \\&= 30 - 17z + z^2\end{aligned}$$

$$\text{Hence, } (2 - z) (15 - z) = 30 - 17z + z^2$$

Question 2(v)

Evaluate :

$$(a^2 + 5) (a^2 - 3)$$

Answer

$$\begin{aligned}(a^2 + 5) (a^2 - 3) \\&= a^2 (a^2 - 3) + 5 (a^2 - 3) \\&= a^{(2+2)} - 3a^2 + 5a^2 - 15 \\&= a^4 + 2a^2 - 15\end{aligned}$$

$$\text{Hence, } (a^2 + 5) (a^2 - 3) = a^4 + 2a^2 - 15$$

Question 2(vi)

Evaluate :

$$(4 - ab) (8 + ab)$$

Answer

$$\begin{aligned}(4 - ab) (8 + ab) \\&= 4 (8 + ab) - ab (8 + ab) \\&= 32 + 4ab - 8ab - a^{(1+1)}b^{(1+1)} \\&= 32 - 4ab - a^2b^2\end{aligned}$$

$$\text{Hence, } (4 - ab) (8 + ab) = 32 - 4ab - a^2b^2$$

Question 2(vii)

Evaluate :

$$(5xy - 7) (7xy + 9)$$

Answer

$$\begin{aligned}(5xy - 7) (7xy + 9) \\&= 5xy (7xy + 9) - 7 (7xy + 9) \\&= 35x^{(1+1)}y^{(1+1)} + 45xy - 49xy - 63 \\&= 35x^2y^2 - 4xy - 63\end{aligned}$$

$$\text{Hence, } (5xy - 7) (7xy + 9) = 35x^2y^2 - 4xy - 63$$

Question 2(viii)

Evaluate :

$$(3a^2 - 4b^2)(8a^2 - 3b^2)$$

Answer

$$(3a^2 - 4b^2)(8a^2 - 3b^2)$$

$$= 3a^2(8a^2 - 3b^2) - 4b^2(8a^2 - 3b^2)$$

$$= 24a^{(2+2)} - 9a^2b^2 - 32a^2b^2 + 12b^{(2+2)}$$

$$= 24a^4 - 41a^2b^2 + 12b^4$$

$$\text{Hence, } (3a^2 - 4b^2)(8a^2 - 3b^2) = 24a^4 - 41a^2b^2 + 12b^4$$

Question 3(i)

Find the square of:

$$3x + \frac{2}{y}$$

Answer

Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

$$\left(3x + \frac{2}{y}\right)^2$$

$$= (3x)^2 + 2 \times 3x \times \frac{2}{y} + \left(\frac{2}{y}\right)^2$$

$$= 9x^2 + \frac{2 \times 6x}{y} + \frac{4}{y^2}$$

$$= 9x^2 + \frac{12x}{y} + \frac{4}{y^2}$$

$$\text{Hence, } \left(3x + \frac{2}{y}\right)^2 = 9x^2 + \frac{12x}{y} + \frac{4}{y^2}$$

Question 3(ii)

Find the square of:

$$\frac{5a}{6b} - \frac{6b}{5a}$$

Answer

Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$\begin{aligned} & \left(\frac{5a}{6b} - \frac{6b}{5a}\right)^2 \\ &= \left(\frac{5a}{6b}\right)^2 - 2 \times \frac{5a}{6b} \times \frac{6b}{5a} + \left(\frac{6b}{5a}\right)^2 \\ &= \frac{25a^2}{36b^2} - \frac{2 \times 5a \times 6b}{6b \times 5a} + \frac{36b^2}{25a^2} \\ &= \frac{25a^2}{36b^2} - \frac{60ab}{30ab} + \frac{36b^2}{25a^2} \\ &= \frac{25a^2}{36b^2} - 2 + \frac{36b^2}{25a^2} \end{aligned}$$

Hence, $\left(\frac{5a}{6b} - \frac{6b}{5a}\right)^2 = \frac{25a^2}{36b^2} - 2 + \frac{36b^2}{25a^2}$

Question 3(iii)

Find the square of:

$$2m^2 - \frac{2}{3}n^2$$

Answer

Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$\left(2m^2 - \frac{2}{3}n^2\right)^2$$

$$= (2m^2)^2 - 2 \times 2m^2 \times \frac{2}{3}n^2 + \left(\frac{2}{3}n^2\right)^2$$

$$= 4m^4 - \frac{2 \times 2m^2 \times 2n^2}{3} + \frac{4}{9}n^4$$

$$= 4m^4 - \frac{8}{3}m^2n^2 + \frac{4}{9}n^4$$

Hence, $\left(2m^2 - \frac{2}{3}n^2\right)^2 = 4m^4 - \frac{8}{3}m^2n^2 + \frac{4}{9}n^4$

Question 3(iv)

Find the square of:

$$5x + \frac{1}{5x}$$

Answer

Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

$$\left(5x + \frac{1}{5x}\right)^2$$

$$= (5x)^2 + 2 \times 5x \times \frac{1}{5x} + \left(\frac{1}{5x}\right)^2$$

$$= 25x^2 + \frac{10x}{5x} + \frac{1}{25x^2}$$

$$= 25x^2 + 2 + \frac{1}{25x^2}$$

$$\text{Hence, } \left(5x + \frac{1}{5x}\right)^2 = 25x^2 + 2 + \frac{1}{25x^2}$$

Question 3(v)

Find the square of:

$$8x + \frac{3}{2}y$$

Answer

Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

$$\begin{aligned}& \left(8x + \frac{3}{2}y\right)^2 \\&= (8x)^2 + 2 \times 8x \times \frac{3}{2}y + \left(\frac{3}{2}y\right)^2 \\&= 64x^2 + \frac{48xy}{2} + \frac{9}{4}y^2 \\&= 64x^2 + 24xy + \frac{9}{4}y^2 \\&\text{Hence, } \left(8x + \frac{3}{2}y\right)^2 = 64x^2 + 24xy + \frac{9}{4}y^2\end{aligned}$$

Question 3(vi)

Find the square of:

607

Answer

Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

$$607^2$$

$$= (600 + 7)^2$$

$$= (600)^2 + 2 \times 600 \times 7 + (7)^2$$

$$= 3,60,000 + 8,400 + 49$$

$$= 3,68,400 + 49$$

$$= 3,68,449$$

$$\text{Hence, } 607^2 = 3,68,449$$

Question 3(vii)

Find the square of:

$$391$$

Answer

Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$391^2$$

$$= (400 - 9)^2$$

$$= (400)^2 - 2 \times 400 \times 9 + (9)^2$$

$$= 1,60,000 - 7,200 + 81$$

$$= 1,52,800 + 81$$

$$= 1,52,881$$

$$\text{Hence, } 391^2 = 1,52,881$$

Question 3(viii)

Find the square of:

9.7

Answer

Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

$$9.7^2$$

$$= (10 - 0.3)^2$$

$$= (10)^2 - 2 \times 10 \times 0.3 + (0.3)^2$$

$$= 100 - 6 + 0.09$$

$$= 94 + 0.09$$

$$= 94.09$$

Hence, $9.7^2 = 94.09$ **Question 4**If $a + \frac{1}{a} = 2$, find :

$$(i) a^2 + \frac{1}{a^2}$$

$$(ii) a^4 + \frac{1}{a^4}$$

Answer

(i) Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$\left(a + \frac{1}{a}\right)^2 = a^2 + 2 \times a \times \frac{1}{a} + \frac{1}{a^2}$$

$$\Rightarrow \left(a + \frac{1}{a}\right)^2 = a^2 + 2 + \frac{1}{a^2}$$

Putting the value $a + \frac{1}{a} = 2$, we get

$$2^2 = a^2 + 2 + \frac{1}{a^2}$$

$$\Rightarrow 4 = a^2 + 2 + \frac{1}{a^2}$$

$$\Rightarrow a^2 + \frac{1}{a^2} = 4 - 2$$

$$\Rightarrow a^2 + \frac{1}{a^2} = 2$$

Hence, the value of $a^2 + \frac{1}{a^2} = 2$.

(ii) Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$\left(a^2 + \frac{1}{a^2}\right)^2 = (a^2)^2 + 2 \times a^2 \times \frac{1}{a^2} + \left(\frac{1}{a^2}\right)^2$$

$$\Rightarrow \left(a^2 + \frac{1}{a^2}\right)^2 = a^4 + 2 + \frac{1}{a^4}$$

Putting the value $a^2 + \frac{1}{a^2} = 2$, we get

$$2^2 = a^4 + 2 + \frac{1}{a^4}$$

$$\Rightarrow 4 = a^4 + 2 + \frac{1}{a^4}$$

$$\Rightarrow a^4 + \frac{1}{a^4} = 4 - 2$$

$$\Rightarrow a^4 + \frac{1}{a^4} = 2$$

Hence, the value of $a^4 + \frac{1}{a^4} = 2$.

Question 5

If $m - \frac{1}{m} = 5$, find:

(i) $m^2 + \frac{1}{m^2}$

(ii) $m^4 + \frac{1}{m^4}$

(iii) $m^2 - \frac{1}{m^2}$

Answer

(i) Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

So,

$$\left(m - \frac{1}{m}\right)^2 = m^2 - 2 \times m \times \frac{1}{m} + \frac{1}{m^2}$$

$$\Rightarrow \left(m - \frac{1}{m}\right)^2 = m^2 - 2 + \frac{1}{m^2}$$

Putting the value $m - \frac{1}{m} = 5$, we get

$$5^2 = m^2 - 2 + \frac{1}{m^2}$$

$$\Rightarrow 25 = m^2 - 2 + \frac{1}{m^2}$$

$$\Rightarrow m^2 + \frac{1}{m^2} = 25 + 2$$

$$\Rightarrow m^2 + \frac{1}{m^2} = 27$$

Hence, the value of $m^2 + \frac{1}{m^2} = 27$.

(ii) Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$\left(m^2 + \frac{1}{m^2}\right)^2 = (m^2)^2 + 2 \times m^2 \times \frac{1}{m^2} + \left(\frac{1}{m^2}\right)^2$$

$$\Rightarrow \left(m^2 + \frac{1}{m^2}\right)^2 = m^4 + 2 + \frac{1}{m^4}$$

Putting the value $m^2 + \frac{1}{m^2} = 27$, we get

$$27^2 = m^4 + 2 + \frac{1}{m^4}$$

$$\Rightarrow 729 = m^4 + 2 + \frac{1}{m^4}$$

$$\Rightarrow m^4 + \frac{1}{m^4} = 729 - 2$$

$$\Rightarrow m^4 + \frac{1}{m^4} = 727$$

Hence, the value of $m^4 + \frac{1}{m^4} = 727$.

(iii) Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$\left(m + \frac{1}{m}\right)^2 = m^2 + 2 \times m \times \frac{1}{m} + \frac{1}{m^2}$$

$$\Rightarrow \left(m + \frac{1}{m}\right)^2 = m^2 + 2 + \frac{1}{m^2}$$

$$\Rightarrow \left(m + \frac{1}{m}\right)^2 = m^2 + \frac{1}{m^2} + 2$$

Putting the value $m^2 + \frac{1}{m^2} = 27$, we get

$$\Rightarrow \left(m + \frac{1}{m}\right)^2 = 27 + 2$$

$$\Rightarrow \left(m + \frac{1}{m}\right)^2 = 29$$

$$\Rightarrow \left(m + \frac{1}{m}\right) = \sqrt{29}$$

Now, using the formula,

$$[\because (x^2 - y^2) = (x - y)(x + y)]$$

So,

$$\left(m^2 - \frac{1}{m^2}\right) = \left(m - \frac{1}{m}\right)\left(m + \frac{1}{m}\right)$$

Putting the value, $\left(m - \frac{1}{m}\right) = 5$ and $\left(m + \frac{1}{m}\right) = \sqrt{29}$

$$\left(m^2 - \frac{1}{m^2}\right) = (5) \times (\sqrt{29})$$

$$m^2 - \frac{1}{m^2} = 5\sqrt{29}$$

Hence, the value of $m^2 - \frac{1}{m^2} = 5\sqrt{29}$.

Question 6

If $a^2 + b^2 = 41$ and $ab = 4$, find:

(i) $a - b$

(ii) $a + b$

Answer

(i) Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

So,

$$(a - b)^2 = a^2 - 2ab + b^2$$

Putting the value, $a^2 + b^2 = 41$ and $ab = 4$

$$\Rightarrow (a - b)^2 = (a^2 + b^2) - 2ab$$

$$\Rightarrow (a - b)^2 = (41) - 2 \times 4$$

$$\Rightarrow (a - b)^2 = 41 - 8$$

$$\Rightarrow (a - b)^2 = 33$$

$$\Rightarrow a - b = \sqrt{33}$$

$$\Rightarrow a - b = \sqrt{33}$$

Hence, the value of $a - b = \sqrt{33}$.

(ii) Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$(a + b)^2 = a^2 + 2ab + b^2$$

Putting the value, $a^2 + b^2 = 41$ and $ab = 4$

$$\Rightarrow (a + b)^2 = (a^2 + b^2) + 2ab$$

$$\Rightarrow (a + b)^2 = (41) + 2 \times 4$$

$$\Rightarrow (a + b)^2 = 41 + 8$$

$$\Rightarrow (a + b)^2 = 49$$

$$\Rightarrow a + b = \sqrt{49}$$

$$\Rightarrow a + b = 7$$

Hence, the value of $a + b = 7$.

Question 7

If $2a + \frac{1}{2a} = 8$, find:

(i) $4a^2 + \frac{1}{4a^2}$

(ii) $16a^4 + \frac{1}{16a^4}$

Answer

(i) Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$\left(2a + \frac{1}{2a}\right)^2 = (2a)^2 + 2 \times 2a \times \frac{1}{2a} + \left(\frac{1}{2a}\right)^2$$

$$\Rightarrow \left(2a + \frac{1}{2a}\right)^2 = 4a^2 + 2 + \frac{1}{4a^2}$$

Putting the value $2a + \frac{1}{2a} = 8$, we get

$$8^2 = 4a^2 + 2 + \frac{1}{4a^2}$$

$$\Rightarrow 64 = 4a^2 + 2 + \frac{1}{4a^2}$$

$$\Rightarrow 4a^2 + \frac{1}{4a^2} = 64 - 2$$

$$\Rightarrow 4a^2 + \frac{1}{4a^2} = 62$$

Hence, the value of $4a^2 + \frac{1}{4a^2}$ is 62.

(ii) Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$\left(4a^2 + \frac{1}{4a^2}\right)^2 = (4a^2)^2 + 2 \times 4a^2 \times \frac{1}{4a^2} + \left(\frac{1}{4a^2}\right)^2$$

$$\Rightarrow \left(4a^2 + \frac{1}{4a^2}\right)^2 = 16a^4 + 2 + \frac{1}{16a^4}$$

Putting the value $4a^2 + \frac{1}{4a^2} = 62$, we get

$$62^2 = 16a^4 + 2 + \frac{1}{16a^4}$$

$$\Rightarrow 3,844 = 16a^4 + 2 + \frac{1}{16a^4}$$

$$\Rightarrow 16a^4 + \frac{1}{16a^4} = 3,844 - 2$$

$$\Rightarrow 16a^4 + \frac{1}{16a^4} = 3,842$$

Hence, the value of $16a^4 + \frac{1}{16a^4}$ is 3,842.

Question 8

If $3x - \frac{1}{3x} = 5$, find:

(i) $9x^2 + \frac{1}{9x^2}$

(ii) $81x^4 + \frac{1}{81x^4}$

Answer

(i) Using the formula,

$$[\because (x - y)^2 = x^2 - 2xy + y^2]$$

So,

$$\left(3x - \frac{1}{3x}\right)^2 = (3x)^2 - 2 \times 3x \times \frac{1}{3x} + \left(\frac{1}{3x}\right)^2$$

$$\Rightarrow \left(3x - \frac{1}{3x}\right)^2 = 9x^2 - 2 + \frac{1}{9x^2}$$

Putting the value $3x - \frac{1}{3x} = 5$, we get

$$5^2 = 9x^2 - 2 + \frac{1}{9x^2}$$

$$\Rightarrow 25 = 9x^2 - 2 + \frac{1}{9x^2}$$

$$\Rightarrow 9x^2 + \frac{1}{9x^2} = 25 + 2$$

$$\Rightarrow 9x^2 + \frac{1}{9x^2} = 27$$

Hence, the value of $9x^2 + \frac{1}{9x^2}$ is 27.

(ii) Using the formula,

$$[\because (x + y)^2 = x^2 + 2xy + y^2]$$

So,

$$\left(9x^2 + \frac{1}{9x^2}\right)^2 = (9x^2)^2 + 2 \times 9x^2 \times \frac{1}{9x^2} + \left(\frac{1}{9x^2}\right)^2$$

$$\Rightarrow \left(9x^2 + \frac{1}{9x^2}\right)^2 = 81x^4 + 2 + \frac{1}{81x^4}$$

Putting the value $9x^2 + \frac{1}{9x^2} = 27$, we get

$$27^2 = 81x^4 + 2 + \frac{1}{81x^4}$$

$$\Rightarrow 729 = 81x^4 + 2 + \frac{1}{81x^4}$$

$$\Rightarrow 81x^4 + \frac{1}{81x^4} = 729 - 2$$

$$\Rightarrow 81x^4 + \frac{1}{81x^4} = 727$$

Hence, the value of $81x^4 + \frac{1}{81x^4}$ is 727.

Question 9(i)

Expand:

$$(3x - 4y + 5z)^2$$

Answer

Using the formula,

$$[\because (x - y + z)^2 = x^2 + y^2 + z^2 - 2xy - 2yz + 2xz]$$

$$\Rightarrow (3x - 4y + 5z)^2 = (3x)^2 + (4y)^2 + (5z)^2 - 2 \times (3x) \times (4y) - 2 \times (4y) \times (5z) + 2 \times (5z) \times (3x)$$

$$= 9x^2 + 16y^2 + 25z^2 - 24xy - 40yz + 30xz$$

Hence, $(3x - 4y + 5z)^2 = 9x^2 + 16y^2 + 25z^2 - 24xy - 40yz + 30xz$.

Question 9(ii)

Expand:

$$(2a - 5b - 4c)^2$$

Answer

Using the formula,

$$[\because (x - y - z)^2 = x^2 + y^2 + z^2 - 2xy + 2yz - 2xz]$$

$$\Rightarrow (2a - 5b - 4c)^2 = (2a)^2 + (5b)^2 + (4c)^2 - 2 \times (2a) \times (5b) + 2 \times (5b) \times (4c) - 2 \times (4c) \times (2a)$$

$$= 4a^2 + 25b^2 + 16c^2 - 20ab + 40bc - 16ca$$

$$\text{Hence, } (2a - 5b - 4c)^2 = 4a^2 + 25b^2 + 16c^2 - 20ab + 40bc - 16ca.$$

Question 9(iii)

Expand:

$$(5x + 3y)^3$$

Answer

Using the formula,

$$[\because (x + y)^3 = x^3 + y^3 + 3x^2y + 3xy^2]$$

So,

$$\Rightarrow (5x + 3y)^3 = (5x)^3 + (3y)^3 + 3 \times (5x)^2 \times (3y) + 3 \times (5x) \times (3y)^2$$

$$= 125x^3 + 27y^3 + 225x^2y + 135xy^2$$

$$\text{Hence, } (5x + 3y)^3 = 125x^3 + 27y^3 + 225x^2y + 135xy^2.$$

Question 9(iv)

Expand:

$$(6a - 7b)^3$$

Answer

Using the formula,

$$[\because (x - y)^3 = x^3 - y^3 - 3x^2y + 3xy^2]$$

So,

$$\Rightarrow (6a - 7b)^3 = (6a)^3 - (7b)^3 - 3 \times (6a)^2 \times (7b) + 3 \times (6a) \times (7b)^2$$

$$= 216a^3 - 343b^3 - 756a^2b + 882ab^2$$

$$\text{Hence, } (6a - 7b)^3 = 216a^3 - 343b^3 - 756a^2b + 882ab^2.$$

Question 10

If $a + b + c = 9$ and $ab + bc + ca = 15$, find: $a^2 + b^2 + c^2$.

Answer

Using the formula,

$$[\because (x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx]$$

So,

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

Putting the value $a + b + c = 9$ and $ab + bc + ca = 15$, we get

$$\Rightarrow 9^2 = a^2 + b^2 + c^2 + 2 \times 15$$

$$\Rightarrow 81 = a^2 + b^2 + c^2 + 30$$

$$\Rightarrow a^2 + b^2 + c^2 = 81 - 30$$

$$\Rightarrow a^2 + b^2 + c^2 = 51$$

Hence, the value of $a^2 + b^2 + c^2$ is 51.

Question 11

If $a + b + c = 11$ and $a^2 + b^2 + c^2 = 81$, find: $ab + bc + ca$.

Answer

Using the formula,

$$[\because (x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx]$$

So,

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$\Rightarrow (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

Putting the value $(a + b + c) = 11$ and $a^2 + b^2 + c^2 = 81$,

$$\Rightarrow (11)^2 = 81 + 2(ab + bc + ca)$$

$$\Rightarrow 121 = 81 + 2(ab + bc + ca)$$

$$\Rightarrow 2(ab + bc + ca) = 121 - 81$$

$$\Rightarrow 2(ab + bc + ca) = 40$$

$$\Rightarrow ab + bc + ca = \frac{40}{2}$$

$$\Rightarrow ab + bc + ca = 20$$

Hence, the value of $(ab + bc + ca)$ is 20.

Question 12

If $3x - 4y = 5$ and $xy = 3$, find: $27x^3 - 64y^3$

Answer

Using the formula,

$$[\because (x - y)^3 = x^3 - y^3 - 3xy(x - y)]$$

So,

$$\Rightarrow (3x - 4y)^3 = (3x)^3 - (4y)^3 - 3 \times 3x \times 4y(3x - 4y)$$

$$\Rightarrow (3x - 4y)^3 = 27x^3 - 64y^3 - 36xy(3x - 4y)$$

Putting the value $(3x - 4y) = 5$ and $xy = 3$, we get

$$\Rightarrow (5)^3 = 27x^3 - 64y^3 - 36 \times 3 \times 5$$

$$\Rightarrow 125 = 27x^3 - 64y^3 - 540$$

$$\Rightarrow 27x^3 - 64y^3 = 125 + 540$$

$$\Rightarrow 27x^3 - 64y^3 = 665$$

Hence, the value of $27x^3 - 64y^3$ is 665.

Question 13

If $a + b = 8$ and $ab = 15$, find: $a^3 + b^3$

Answer

Using the formula,

$$[\because (x + y)^3 = x^3 + y^3 + 3xy(x + y)]$$

So,

$$\Rightarrow (a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

Putting the value $(a + b) = 8$ and $ab = 15$, we get

$$\Rightarrow (8)^3 = a^3 + b^3 + 3 \times 15 \times 8$$

$$\Rightarrow 512 = a^3 + b^3 + 360$$

$$\Rightarrow a^3 + b^3 = 512 - 360$$

$$\Rightarrow a^3 + b^3 = 152$$

Hence, the value of $a^3 + b^3$ is 152.

Question 14

If $3x + 2y = 9$ and $xy = 3$, find: $27x^3 + 8y^3$

Answer

Using the formula,

$$[\because (x + y)^3 = x^3 + y^3 + 3xy(x + y)]$$

So,

$$\Rightarrow (3x + 2y)^3 = (3x)^3 + (2y)^3 + 3 \times 3x \times 2y(3x + 2y)$$

$$\Rightarrow (3x + 2y)^3 = 27x^3 + 8y^3 + 18xy(3x + 2y)$$

Putting the value $(3x + 2y) = 9$ and $xy = 3$, we get

$$\Rightarrow (9)^3 = 27x^3 + 8y^3 + 18 \times 3 \times 9$$

$$\Rightarrow 729 = 27x^3 + 8y^3 + 486$$

$$\Rightarrow 27x^3 + 8y^3 = 729 - 486$$

$$\Rightarrow 27x^3 + 8y^3 = 243$$

Hence, the value of $27x^3 + 8y^3$ is 243.

Question 15

If $5x - 4y = 7$ and $xy = 8$, find: $125x^3 - 64y^3$

Answer

Using the formula,

$$[\because (x - y)^3 = x^3 - y^3 - 3xy(x - y)]$$

So,

$$\Rightarrow (5x - 4y)^3 = (5x)^3 - (4y)^3 - 3 \times 5x \times 4y(5x - 4y)$$

$$\Rightarrow (5x - 4y)^3 = 125x^3 - 64y^3 - 60xy(5x - 4y)$$

Putting the value $(5x - 4y) = 7$ and $xy = 8$, we get

$$\Rightarrow (7)^3 = 125x^3 - 64y^3 - 60 \times 8 \times 7$$

$$\Rightarrow 343 = 125x^3 - 64y^3 - 3,360$$

$$\Rightarrow 125x^3 - 64y^3 = 343 + 3,360$$

$$\Rightarrow 125x^3 - 64y^3 = 3,703$$

Hence, the value of $125x^3 - 64y^3$ is 3,703.

Question 16

The difference between two numbers is 5 and their product is 14. Find the difference between their cubes.

Answer

Let the two numbers be x and y .

So,

$$x - y = 5$$

And,

$$xy = 14$$

Using the formula,

$$[\because (x - y)^3 = x^3 - y^3 - 3xy(x - y)]$$

So,

$$\Rightarrow (x - y)^3 = x^3 - y^3 - 3xy(x - y)$$

Putting the value $(x - y) = 5$ and $xy = 14$, we get

$$\Rightarrow (5)^3 = x^3 - y^3 - 3 \times 14 \times 5$$

$$\Rightarrow 125 = x^3 - y^3 - 210$$

$$\Rightarrow x^3 - y^3 = 125 + 210$$

$$\Rightarrow x^3 - y^3 = 335$$

Hence, the difference between their cubes is 335.

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